MULTI-OBJECTIVE OPTIMIZATION MODEL FOR HOSPITAL EMERGENCY GENERATOR MANAGEMENT

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Abstract

This report presents a comprehensive multi-objective optimization model for managing emergency generators in hospital facilities during power outages. The model employs Goal Programming to balance service utility maximization with equity considerations, while incorporating investment decisions for generator acquisition. Through extensive scenario analysis, we evaluate the impact of budget constraints, blackout duration, and objective weighting on optimal resource allocation and service delivery.

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1 Executive Summary

Emergency preparedness is critical for hospital operations, particularly regarding power supply continuity. During blackout events, hospitals must maintain essential services using backup generator systems while managing limited fuel resources and potentially expanding generator capacity through strategic investments.

This report presents a sophisticated optimization model that addresses three interconnected decision problems: (1) optimal investment in additional generator capacity subject to budget constraints, (2) dynamic activation scheduling of available generators, and (3) real-time service level management balancing criticality priorities with equity considerations.

1.1 Key Features of the Model

Our approach incorporates several novel elements that distinguish it from traditional emergency management models:

Multi-Unit Investment Framework: Unlike models that assume fixed infrastructure, we explicitly model the decision to purchase additional generator units across different capacity tiers (Large, Medium, Small), each with distinct cost-performance characteristics.

Goal Programming Architecture: We employ a weighted goal programming formulation that jointly addresses multiple, potentially conflicting objectives: (i) maximizing the aggregate service utility delivered over the planning horizon (weighted by each service's criticality), (ii) maximizing equity across services through a maximin-type criterion that promotes balanced service provision, and (iii) extending as much as possible over time the period during which the hospital remains operational (i.e., keeping all critical services at the required operational level for the longest attainable duration), thereby ensuring the hospital can be kept alive throughout the blackout.

Partial Service Capability: Rather than binary on/off decisions for services, we allow continuous service levels between 0 and 1, enabling nuanced resource allocation and graceful degradation strategies.

2 Introduction

2.1 Problem Context and Motivation

Healthcare facilities face unique challenges during power outages. Unlike commercial or residential settings where temporary power loss represents an inconvenience, hospitals provide life-sustaining services where electrical interruption can have immediate and severe consequences. Operating rooms require stable power for surgical equipment and life support systems. Intensive Care Units (UCIs) depend on continuous electrical supply for ventilators,

monitors, and infusion pumps. Emergency departments must maintain imaging capabilities, laboratory services, and patient monitoring regardless of grid status.

The criticality of uninterrupted power has driven regulatory requirements for backup generation capacity in hospitals. However, designing and operating these systems involves complex tradeoffs. Generator capacity represents significant capital investment, with larger industrial generators costing of thousands dollars. Fuel storage infrastructure adds additional costs and spatial requirements. During actual outage events, operational decisions become equally complex as limited fuel reserves must be allocated across competing demands while maintaining minimum safety standards for critical services.

Recent events have highlighted the vulnerability of power infrastructure. Natural disasters such as hurricanes, earthquakes, and severe storms can disrupt electrical grids for extended periods. Infrastructure failures, cyberattacks, and cascading grid failures represent additional risks. Climate change projections suggest increasing frequency and severity of weather events that threaten power reliability. These trends underscore the importance of robust emergency power planning and operational capabilities.

2.2 Research Objectives

This research develops and demonstrates a comprehensive optimization framework for hospital emergency power management that addresses three interconnected decision problems:

Strategic Investment Planning: Given budget constraints and cost-capacity tradeoffs across different generator types, determine the optimal mix of generator purchases to maximize emergency response capability.

Tactical Activation Scheduling: For a given generator fleet (existing plus newly purchased units), determine when to start each generator considering their specific features, fuel limitations, and evolving service demands.

Operational Service Management: At each time period during a blackout, determine service levels for all hospital departments balancing multiple objectives including aggregate utility maximization, equity across services, and continuity of critical services.

Our model integrates these three decision layers within a unified optimization framework, enabling holistic analysis of investment-operations tradeoffs that are typically addressed separately in practice.

3 Mathematical Formulation

This section presents the complete mathematical model for hospital emergency generator management. We employ standard optimization modeling conventions with clearly defined sets, parameters, decision variables, objective function, and constraints.

3.1 Sets and Indices

The model structure is built on four fundamental sets that capture the essential elements of the system:

- $T = \{1, 2, \ldots, T_{\max}\}$: Set of discrete time periods spanning the blackout horizon. Each period represents one hour. We adopt the convention that $t = 1$ corresponds to the period when the blackout begins, enabling modeling of immediate response actions.
- $S = \{s_1, s_2, \ldots, s_{|S|}\}$: Set of hospital services and departments requiring electrical power. This includes clinical services (UCI, Operating Rooms, Emergency Room) as well as support services (Laboratory, Administration, Pharmacy). Each service is treated as a distinct entity with specific power requirements and criticality.
- $S_{\text{crit}} \subseteq S$: Subset of critical services whose operation is essential for minimum hospital functionality. This designation determines which services receive priority treatment and influences the definition of "hospital operational" status. The specific composition of S_{crit} is a policy parameter typically defined by medical staff and regulatory requirements.
- $G = \{g_1, g_2, \ldots, g_{|G|}\}$: Set of generator types available for the hospital's emergency power system. Each type represents a distinct class of generator characterized by capacity, fuel efficiency, cost, and operational parameters. In our analysis, we consider three types: Large (500 kW, 5000 kWh capacity), Medium (150 kW, 1500 kWh capacity), and Small (100 kW, 600 kWh capacity).
- $G_{\text{cand}} \subseteq G$: Subset of candidate (optional) generator types that can be acquired/installed as an investment decision if the existing emergency capacity is insufficient. Elements of G_{cand} are associated with fixed investment costs and can only contribute power and energy to the system if selected.

These sets define the dimensionality of the decision space and the granularity of the model. The size of set T determines the temporal resolution of the plan, while the composition of sets S and G reflects the specific hospital facility being modeled.

3.2 Parameters

Model parameters represent fixed inputs including technical specifications, cost data, and policy parameters. These values have been determined through data collection, research, engineering specifications, and policy decisions to test different scenarios.

3.2.1 Generator Technical Specifications

For each generator type $g \in G$, we define:

- $P_g^{\max}$ [kW]: Maximum instantaneous power output capacity. This represents the rated output of the generator under standard operating conditions and determines the maximum load that can be served by active generators of type g in any single period.
- $E_g^{\max}$ [kWh]: Total fuel energy capacity for a new generator of type g. This parameter combines tank capacity with generator fuel efficiency to express total energy production capability from a full fuel load. It represents the integrated energy output possible over the generator's operation before refueling is required.
- E_g^{initial} [kWh]: Initial fuel energy available in existing generators of type g. This captures the realistic scenario where existing generators may not have full tanks at the moment of blackout occurrence.

3.2.2 Generator Economic Parameters

- Cost_g [$]: Purchase price for one unit of generator type g. This represents the capital cost of acquiring and installing a new generator, including the basic unit cost plus installation, connection to the electrical distribution system, and commissioning.
- Budget [$]: Total budget available for generator purchases during the planning period. This constraint reflects capital budget limitations and represents one of the fundamental tradeoffs in the model: how to optimally allocate limited financial resources across generator types with different cost-performance characteristics.
- N_g^{exist} [units]: Number of existing generators of type g already owned by the hospital. These units are available for use during emergencies without requiring additional investment, though they are subject to their initial fuel levels E_g^{initial}.

3.2.3 Service Demand and Criticality

For each service $s \in S$:

- Dem_s [kW]: Electrical power demand when service s operates at full capacity. This represents the continuous power draw required to maintain normal operations for the service, including equipment, lighting, climate control, and other electrical loads.
- W_s [dimensionless]: Criticality weight or importance factor for service s. This parameter quantifies the relative value of maintaining service s and typically reflects clinical priorities, safety requirements, and regulatory mandates. Services like UCI and Operating Rooms have high weights, while administrative services have lower weights. These weights are typically established through clinical consensus and policy review.

3.2.4 Goal Programming Parameters

- $\lambda_1 \in [0, 1]$: Weight assigned to the utility maximization objective. This parameter determines the relative importance of maximizing aggregate service delivery weighted by criticality.
- $\lambda_2 \in [0, 1]$: Weight assigned to the equity objective. This parameter determines the relative importance of ensuring balanced service levels across departments. By convention, we typically set $\lambda_2 = 1 - \lambda_1$ to maintain normalized weights, though this is not strictly required.
- K [dimensionless]: Normalization factor for the maximin equity term. This scaling parameter ensures that the utility and equity objectives are measured on comparable scales, enabling meaningful interpretation of the weights λ_1 and λ_2. We compute K such that perfect equity (all services at maximum level) produces a contribution to the objective comparable to maximum utility.
- α [dimensionless]: Urgency decay rate (per period) used to discount service delivery over time, so that earlier periods are prioritized during the blackout. It is calibrated to achieve a prescribed residual weight in the final period.

3.2.5 Policy Parameters

- $L^{\min}$ [dimensionless]: Minimum service level required for critical services when the hospital is considered operational. This threshold (we have used typically 0.10) ensures that when critical services are "active," they maintain at least minimal functionality rather than just token operation.
- $\Delta^{\max}$ [dimensionless]: Maximum allowable gap between service levels of any two critical services in the same period. This equity constraint (we have used typically 0.20) prevents extreme disparities where one critical service operates at full capacity while another barely functions, promoting solidarity among essential services.

3.3 Decision Variables

The model employs several classes of decision variables representing investment decisions, generator operations, and service levels.

3.3.1 Investment Decisions

$$z_g^{\text{inv}} \in \mathbb{Z}_+ \quad \forall g \in G \tag{1}$$

The integer variable z_g^{inv} represents the number of new generators of type g to purchase. This is the strategic investment decision that determines the expanded capacity of the emergency

power system. The optimal values balance the tradeoff between acquisition cost and the enhanced operational flexibility provided by additional capacity.

3.3.2 Generator Operations

$$s_{g,t} \in \mathbb{Z}_+ \quad \forall g \in G, \forall t \in T \tag{2}$$
$$a_{g,t} \in \mathbb{Z}_+ \quad \forall g \in G, \forall t \in T \tag{3}$$
$$u_{g,t} \in \mathbb{R}_+ \quad \forall g \in G, \forall t \in T \tag{4}$$

These variables capture the operational state and output of generators:

$s_{g,t}$ represents the number of generators of type g that receive startup commands at the beginning of period t. This variable models the decision of when to activate generators.

$a_{g,t}$ indicates the number of generators of type g that are fully operational (online and available to generate power) during period t. This variable is derived from startup decisions via constraints.

$u_{g,t}$ specifies the actual energy generated by all generators of type g during period t measured in kWh. This continuous variable allows the model to determine optimal generation levels considering fuel constraints and demand requirements, enabling partial utilization of available capacity when appropriate.

3.3.3 Service Level Variables

$$x_{s,t} \in [0, 1] \quad \forall s \in S, \forall t \in T \tag{5}$$
$$\tag{6}$$

These variables characterize service delivery:

$x_{s,t}$ represents the service level or fraction of normal operations for service s during period t. A value of 1.0 indicates full operation at design capacity, 0.5 indicates operation at half capacity, and 0.0 indicates the service is offline. This continuous formulation enables graceful degradation and nuanced resource allocation.

3.3.4 Auxiliary Variables

$$z^{\min} \in \mathbb{R}_+ \tag{7}$$

The variable $z^{\min}$ represents the minimum normalized service level across all services and all time periods, defined as:

$$z^{\min} = \min_{s \in S, t \in T} \left\{ \frac{x_{s,t}}{W_s} \right\} \tag{8}$$

This maximin variable operationalizes the equity objective by identifying the worst-served service (accounting for criticality weights). Maximizing $z^{\min}$ encourages solutions where even the lowest-priority services receive reasonable attention, preventing extreme resource concentration on critical services at the expense of abandoning others entirely.

3.4 Objective Function

The objective function employs goal programming to balance two potentially conflicting goals: utility maximization and equity promotion.

3.4.1 Goal 1: Weighted Utility Maximization

The utility component aggregates service delivery weighted by criticality and time:

$$G_1 = \sum_{t \in T} \sum_{s \in S} \frac{W_s \cdot x_{s,t}}{(1+\alpha)^{t-1}} \tag{9}$$

This term accumulates "utility points" for each service-period combination, where the contribution is proportional to the service's criticality weight W_s and its operational level $x_{s,t}$. The temporal discount factor $(1+\alpha)^{-(t-1)}$ introduces urgency, making service delivery more valuable in later periods when the crisis has persisted longer and alternative options may be more limited.

For example, operating an UCI (assume $W_{\text{UCI}} = 100$) at 80% capacity for 24 hours accumulates substantial utility, while operating administrative services (assume $W_{\text{Admin}} = 5$) at the same level contributes much less to G_1.

3.4.2 Goal 2: Maximin Equity

The equity component focuses on the worst-served department:

$$G_2 = K \cdot z^{\min} \tag{10}$$

where $z^{\min}$ is constrained by:

$$z^{\min} \leq \frac{x_{s,t}}{W_s} \quad \forall s \in S, \forall t \in T \tag{11}$$

This maximin formulation ensures that optimization considers the service with the lowest normalized service level. The normalization by W_s accounts for the fact that we expect critical services to operate at higher absolute levels, but we still want to ensure that even lower-priority services receive proportionate attention.

The scaling factor K ensures that G_2 has comparable magnitude to G_1, making the weights λ_1 and λ_2 meaningful indicators of preference rather than arbitrary scaling factors.

3.4.3 Composite Objective

The overall objective combines both goals:

$$\max \quad Z = \lambda_1 \cdot G_1 + \lambda_2 \cdot G_2 \tag{12}$$

$$\max \quad Z = \lambda_1 \left(\sum_{t \in T} \sum_{s \in S} \frac{W_s \cdot x_{s,t}}{(1+\alpha)^{t-1}} \right) + \lambda_2 \left(K \cdot z^{\min} \right) \tag{13}$$

The weights λ_1 and λ_2 provide explicit control over the utility-equity tradeoff:

When $\lambda_1 \gg \lambda_2$, the model prioritizes aggregate utility, resulting in solutions that focus resources on the highest-criticality services even if this creates large disparities in service levels.

When $\lambda_2 \gg \lambda_1$, the model prioritizes equity, resulting in more uniform service levels across departments even if this means lower absolute levels for critical services.

When $\lambda_1 \approx \lambda_2$, the model seeks balanced solutions that provide good aggregate utility while avoiding extreme inequities.

3.5 Constraints

The model's feasibility region is defined by constraints that ensure technical feasibility, resource limitations, and policy requirements.

3.5.1 Budget Constraint

Investment decisions must respect the available budget:

$$\sum_{g \in G} \text{Cost}_g \cdot z_g^{\text{inv}} \leq \text{Budget} \tag{14}$$

This constraint prevents over-commitment of financial resources. The left-hand side computes total capital expenditure across all generator types, which cannot exceed the allocated

budget.

3.5.2 Generator Availability Constraints

The number of active generators cannot exceed the total inventory:

$$a_{g,t} \leq N_g^{\text{exist}} + z_g^{\text{inv}} \quad \forall g \in G, \forall t \in T \tag{15}$$

This ensures operational decisions respect physical availability. The right-hand side represents total inventory (existing plus purchased units), while the left-hand side represents generators in operation. This constraint prevents the model from "creating" non-existent capacity.

3.5.3 Power Output Constraints

Generated power is limited by generator capacity:

$$u_{g,t} \leq P_g^{\max} \cdot a_{g,t} \quad \forall g \in G, \forall t \in T \tag{16}$$

This constraint links operational status to generation capacity. When $a_{g,t} = 0$ (no generators of type g are online), we must have $u_{g,t} = 0$ (no generation). When $a_{g,t} > 0$, generation is limited by the product of unit capacity $P_g^{\max}$ and the number of active units.

3.5.4 Fuel Capacity Constraints

Total energy generation over the horizon cannot exceed available fuel:

$$\sum_{t \in T} u_{g,t} \leq \left(E_g^{\text{initial}} \cdot N_g^{\text{exist}}\right) + \left(E_g^{\max} \cdot z_g^{\text{inv}}\right) \quad \forall g \in G \tag{17}$$

This constraint prevents exceeding fuel capacity. The right-hand side computes total fuel energy available: existing generators start with E_g^{initial} energy each, while newly purchased generators have full tanks $E_g^{\max}$. The left-hand side accumulates all energy produced by type g generators across all periods.

This constraint couples investment and operational decisions: purchasing more generators not only adds capacity but also adds fuel reserves, which may be the binding constraint in long blackouts.

3.5.5 Energy Balance Constraints

Supply must equal demand in every period:

$$\sum_{g \in G} u_{g,t} = \sum_{s \in S} \text{Dem}_s \cdot x_{s,t} \quad \forall t \in T \tag{18}$$

This equality constraint ensures that power generation exactly matches consumption. We use equality (rather than $\geq$) because the model operates on a demand-driven basis: generators produce only what is needed, without excess generation or storage capabilities.

The left side aggregates generation across all generator types, while the right side computes total demand as the sum of each service's full demand Dem_s scaled by its operational level $x_{s,t}$.

3.5.6 Generator Startup Dynamics

Startups are recorded whenever the number of active units increases.

$$\text{start}_{g,1} \geq a_{g,1} \quad \forall g \in G \tag{19}$$

$$\text{start}_{g,t} \geq a_{g,t} - a_{g,t-1} \quad \forall g \in G,\ \forall t \in T,\ t > 1 \tag{20}$$

These constraints ensure that any activation at $t = 1$ is counted as a startup, and for $t > 1$ the startup variable is triggered only when switching from off to on. This enables limiting and/or penalizing startups consistently with generator operations.

3.5.7 Total Start Limit

$$\sum_{t \in T} \text{start}_{g,t} \leq 5 \quad \forall g \in G \tag{21}$$

This constraint limits the total number of startups allowed for each generator type over the entire planning horizon, preventing excessive cycling and ensuring operational feasibility.

3.5.8 Critical Service Continuity

Critical services cannot experience abrupt degradation:

$$x_{s,t} \leq x_{s,t-1} \quad \forall s \in S_{\text{crit}},\ \forall t \in T : t > 1 \tag{22}$$

This constraint requires that critical service levels can only decrease or remain constant over time, never increase. This reflects the practical reality that once a critical service has been degraded, ramping it back up may be difficult or impossible during an ongoing emergency.

3.5.9 Equity Gap Constraints

Service levels among critical services cannot diverge excessively:

$$x_{s_1,t} - x_{s_2,t} \leq \Delta^{\max} \quad \forall s_1, s_2 \in S_{\text{crit}}, \forall t \in T \tag{23}$$

These constraints ensure that no two critical services have service levels differing by more than $\Delta^{\max}$ (typically 0.20 or 20%). This promotes solidarity among critical services, preventing solutions where, for example, the UCI operates at 100% while the ER operates at 40% in the same period. Combined with the continuity constraints, these gap constraints encourage coordinated degradation of critical services rather than abandoning some while fully supporting others.

3.5.10 Maximin Definition Constraints

The maximin variable must be bounded by all normalized service levels:

$$z^{\min} \leq \frac{x_{s,t}}{W_s} \quad \forall s \in S, \forall t \in T \tag{24}$$

These constraints ensure that $z^{\min}$ equals the minimum of all normalized service levels. The optimization will push $z^{\min}$ as high as possible (since it appears positively in the objective), but these constraints prevent it from exceeding any actual normalized service level. At optimality, $z^{\min}$ will equal the smallest $x_{s,t}/W_s$ across all service-time combinations, implementing the maximin equity criterion.

3.5.11 Variable Domains

Finally, we specify the domains for all decision variables:

$$z_g^{\text{inv}} \in \mathbb{Z}_+ \quad \forall g \in G \tag{25}$$

$$s_{g,t}, a_{g,t} \in \mathbb{Z}_+ \quad \forall g \in G, \forall t \in T \tag{26}$$

$$u_{g,t} \in \mathbb{R}_+ \quad \forall g \in G, \forall t \in T \tag{27}$$

$$x_{s,t} \in [0,1] \quad \forall s \in S, \forall t \in T \tag{28}$$

$$z^{\min} \in \mathbb{R}_+ \tag{29}$$

Investment, startup, and availability variables are non-negative integers representing counts of physical units. Generation variables are non-negative continuous values representing energy quantities. Service levels are continuous in the unit interval. Hospital operational status is binary. The maximin variable is a non-negative continuous auxiliary variable.

4 Sensitivity Analysis: Immediacy vs. Persistence (16 Hours)

In this section, we analyze the system's behavior during a short-term crisis (16 hours) with a fixed budget of 20,000. We compare two opposing strategies defined by the objective function weights:

- **High Immediacy Priority ($\lambda_1 = 0.9$):** Prioritizes maximum instantaneous power delivery to satisfy critical demands early on.
- **High Persistence Priority ($\lambda_1 = 0.1, \lambda_2 = 0.9$):** Prioritizes extending service duration and minimizing inequality between departments.

4.1 High Utility Priority ($\lambda_1 = 0.9$) - 16 Hours

With a focus on maximizing total utility ($\lambda_1 = 0.9$), the model exhibits an aggressive strategy of "Immediacy."

Observations:

- **Service Levels (SCADA):** As shown in Figure 1, critical services such as the UCI,Operating Rooms and ER are powered at maximum capacity (100%) from the very first hour. However, this high performance comes at a cost: support services like the Pharmacy and Admin experience early shutdowns or operate at minimal levels to divert resources to the critical units.
- **Fuel Depletion:** The fuel breakdown in Figure 2 reveals a steep depletion curve. Generators are run at high loads to meet the intense initial demand, exhausting the fuel reserves rapidly. This confirms the model's preference for gaining utility points "now" rather than saving them for "later."

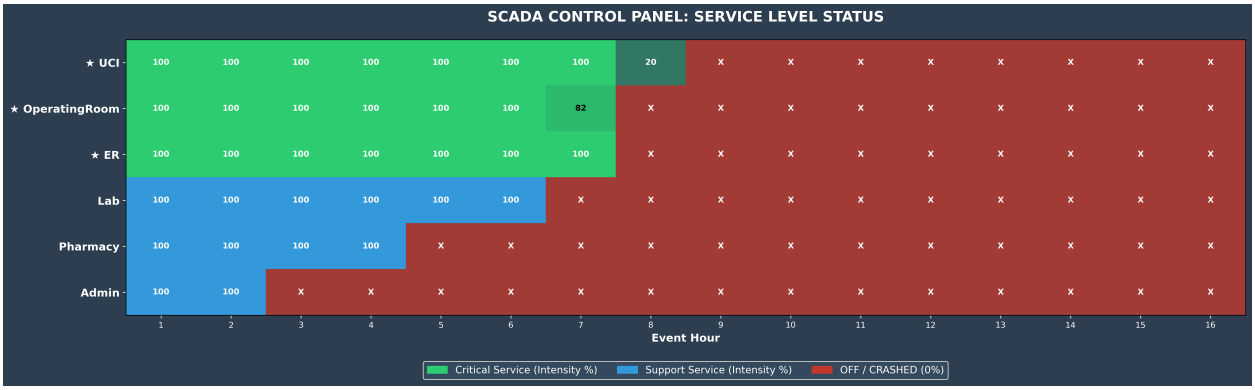


Figure 1: Service Levels (SCADA) - Utility Focused ($\lambda_1 = 0.9$, 16h)

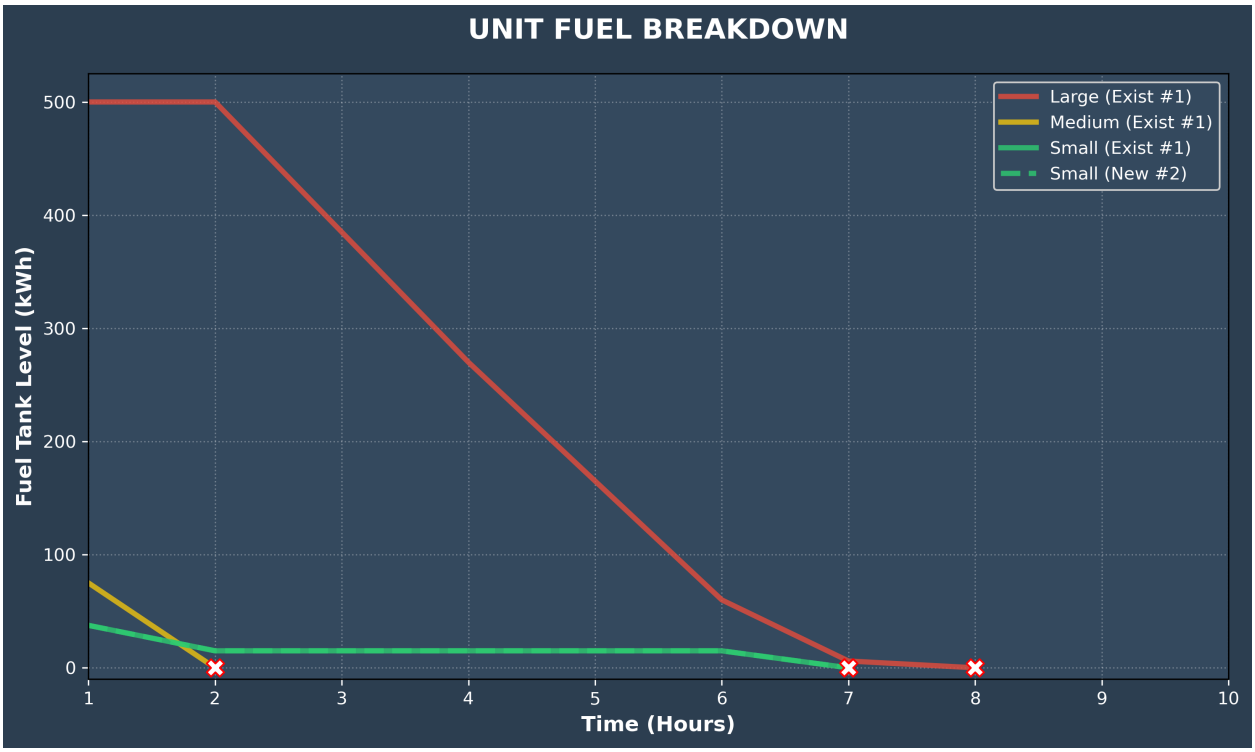


Figure 2: Fuel Depletion - Utility Focused ($\lambda_1 = 0.9$, 16h)

4.2 High Equity Priority ($\lambda_1 = 0.1$) - 16 Hours

In this configuration ($\lambda_1 = 0.1, \lambda_2 = 0.9$), the system shifts towards a strategy of "Persistence" and solidarity.

Observations:

- **Service Levels (SCADA):** Figure 3 shows a markedly different pattern. The service lines are grouped closer together. The model prevents any single service from dropping to zero, even if it means capping the performance of critical units (e.g., UCI runs at 60% instead of 100%). This ensures that all departments remain operational throughout the 16-hour window.
- **Fuel Depletion:** The fuel consumption in Figure 4 is more gradual and linear. By avoiding peak loads, the system conserves fuel, ensuring that the generators can remain active until the very end of the event, albeit at partial capacity. This demonstrates the model's effort to maximize the minimum service level over time.

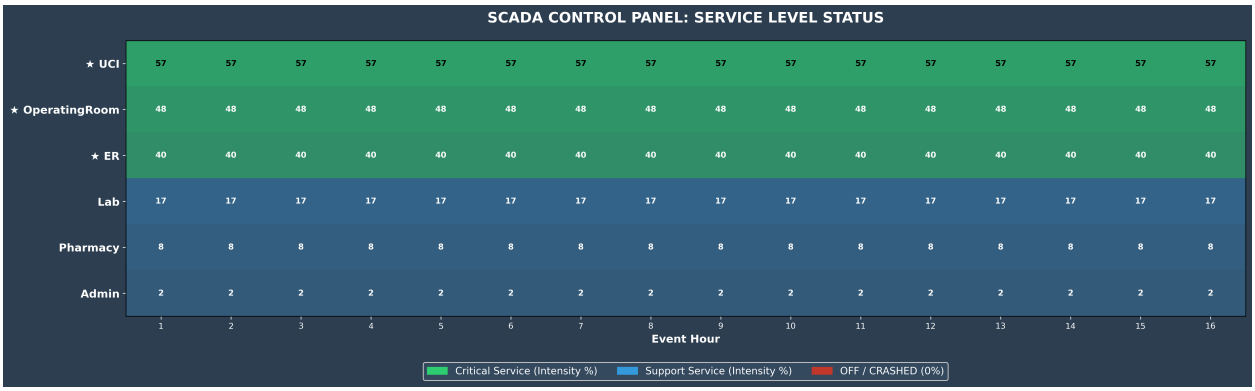


Figure 3: Service Levels (SCADA) - Equity Focused ($\lambda_1 = 0.1$, 16h)

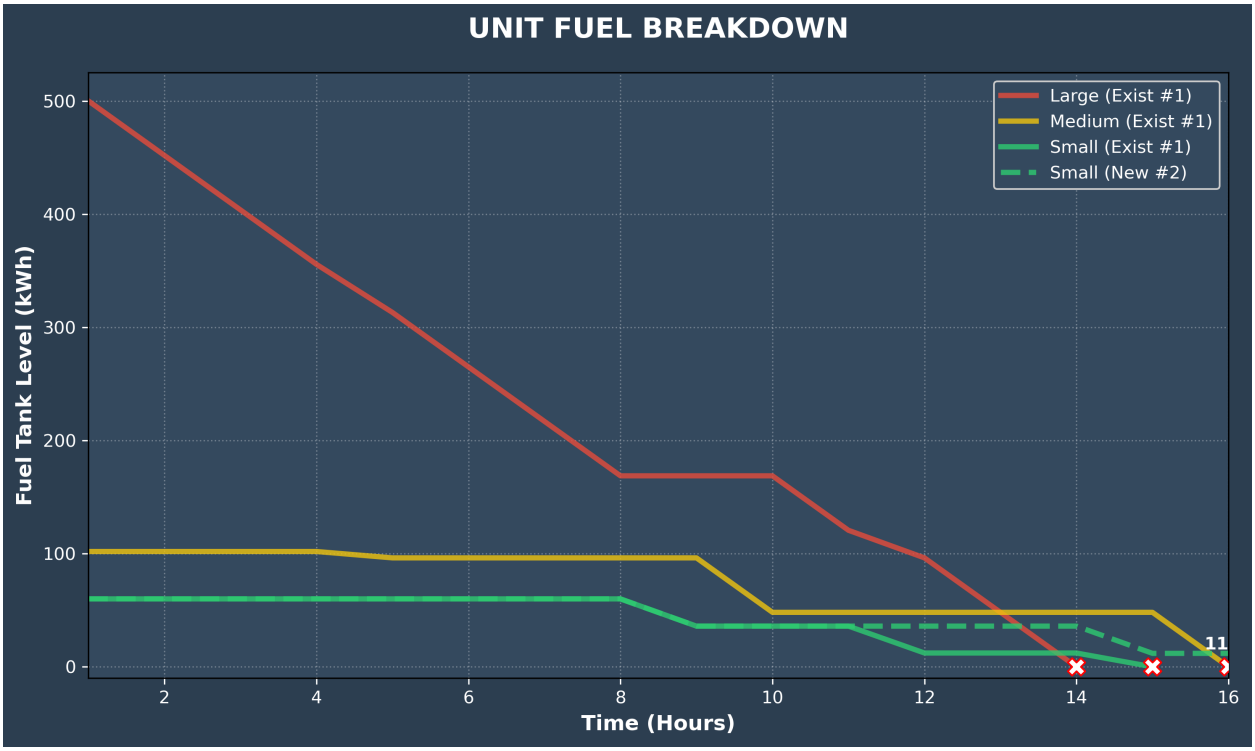


Figure 4: Fuel Depletion - Equity Focused ($\lambda_1 = 0.1$, 16h)

5 Sensitivity Analysis: Utility vs. Equity (24 Hours)

We extend the optimization horizon to 24 hours. The general dynamics observed in the 16-hour scenario remain consistent, but the extended duration introduces subtle yet significant adjustments in resource allocation.

5.1 High Utility Priority ($\lambda_1 = 0.9$) - 24 Hours

With a focus on maximizing total utility, the "Immediacy" strategy persists. However, the extended timeframe forces the model to make slightly harder sacrifices.

Observations:

- **Service Levels (SCADA):** Similar to the 16-hour case, critical services are prioritized. A key difference observed in Figure 5 is the complete suppression of the **Admin** service. By cutting this non-critical load entirely, the system diverts that energy to sustain the high-intensity performance of critical sectors (UCI, OR, ER).
- **Fuel Depletion:** The fuel usage pattern remains aggressive (Figure 6), confirming that the core strategy is unchanged: burn resources early to maximize utility points before the urgency rate devalues them.

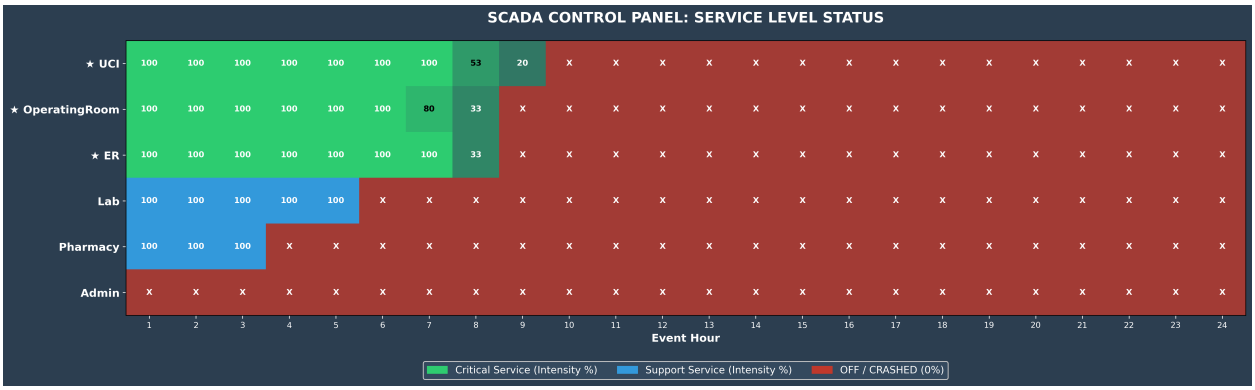


Figure 5: Service Levels (SCADA) - Utility Focused ($\lambda_1 = 0.9$, 24h)

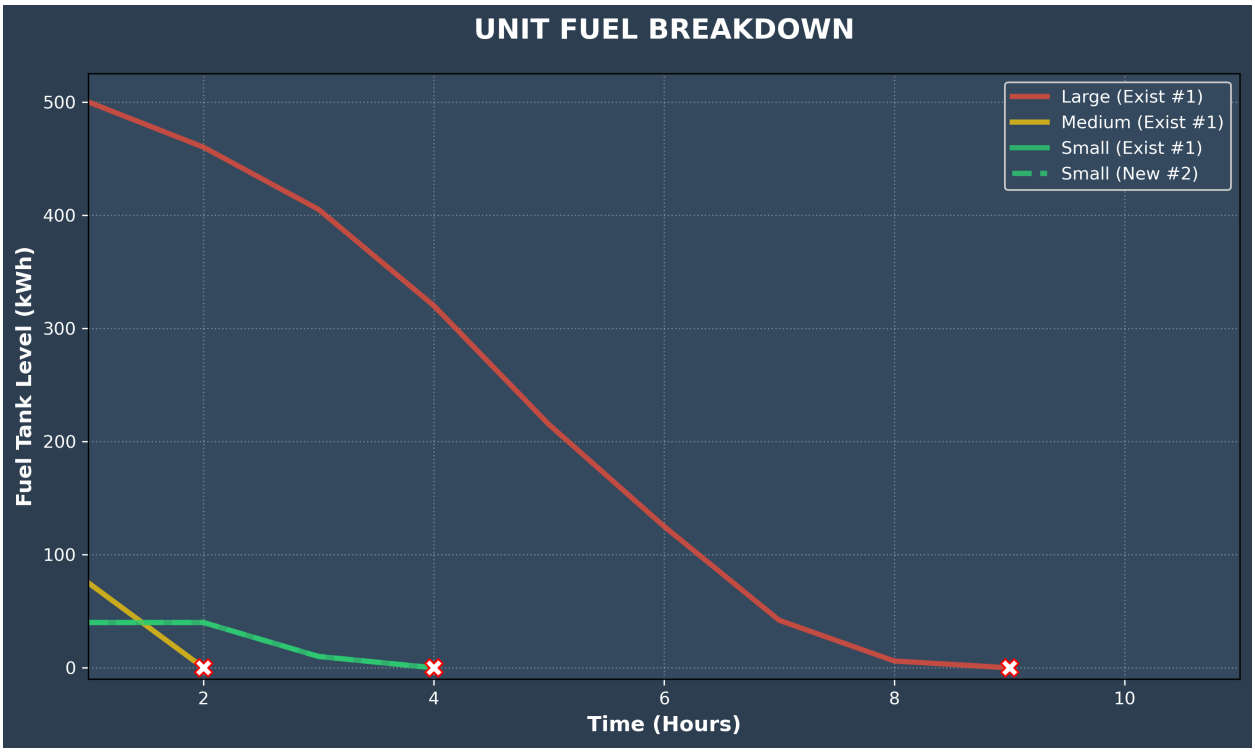


Figure 6: Fuel Depletion - Utility Focused ($\lambda_1 = 0.9$, 24h)

5.2 High Equity Priority ($\lambda_1 = 0.1$) - 24 Hours

Under high equity constraints ($\lambda_2 = 0.9$), the "Persistence" strategy is put to a greater test.

Observations:

- **Service Levels (SCADA):** The behavior mirrors the 16-hour scenario, maintaining a uniform distribution of service levels (Figure 7). However, to accommodate the 50% increase in duration (from 16 to 24 hours), the **average service level is lower**. The system automatically recalibrates to a lower "survival mode" intensity to stretch the same fuel reserves over a longer period.
- **Fuel Depletion:** The generators continue to deplete gradually (Figure 8), but the rate of consumption is adjusted downwards. This confirms the model's predictive capability to ration resources effectively based on the total event horizon.

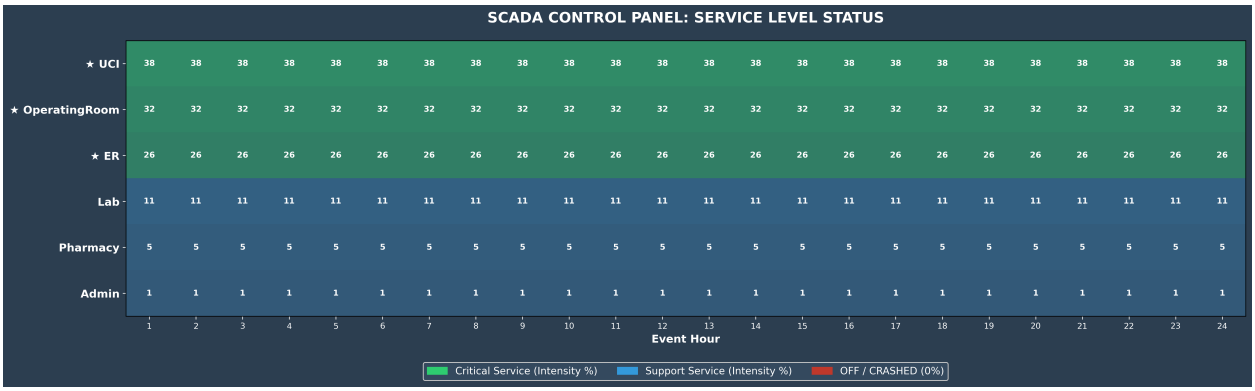


Figure 7: Service Levels (SCADA) - Equity Focused ($\lambda_1 = 0.1$, 24h)

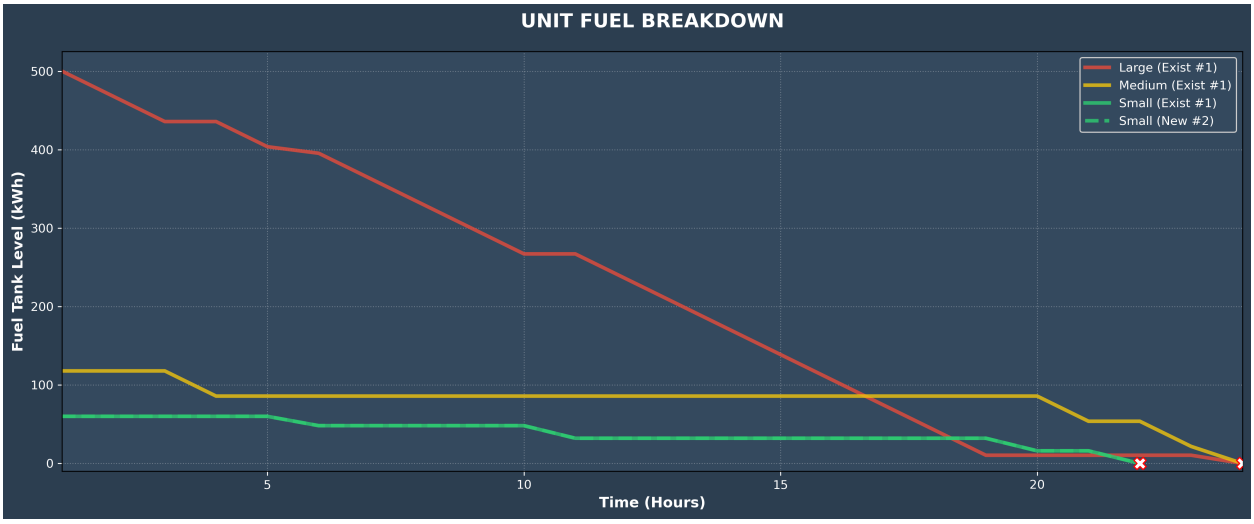


Figure 8: Fuel Depletion - Equity Focused ($\lambda_1 = 0.1$, 24h)

6 Budget Impact Analysis (Balanced Strategy - 24 Hours)

In this section, we analyze a "Balanced" scenario where the weights are equally distributed ($\lambda_1 = 0.5, \lambda_2 = 0.5$). This configuration seeks a middle ground between the aggressive utility maximization of Scenario 1 and the conservative persistence of Scenario 2.

We fix the duration to 24 hours and compare the system's performance under two budget constraints: a standard budget of \$20,000 versus a doubled budget of \$40,000.

6.1 Standard Budget (\$20,000)

With a limited budget, the model is restricted to optimizing the existing inventory and purchasing only minor assets (e.g., a Small generator).

Observations:

- **Balanced Behavior:** As expected, the system exhibits a hybrid behavior. It boosts critical services during the first 3 hours (capitalizing on the high urgency rate for utility points) but quickly transitions to a sustained "maintenance mode" to satisfy the equity constraint over the remaining time. It is neither as aggressive as the pure utility model nor as flat as the pure equity model.
- **Fuel Depletion:** The fuel depletion curve (Figure 10) shows a steady decline. The limited generation capacity forces the system to run generators at high loads relative to their size to meet the balanced demand, leaving little margin for error.

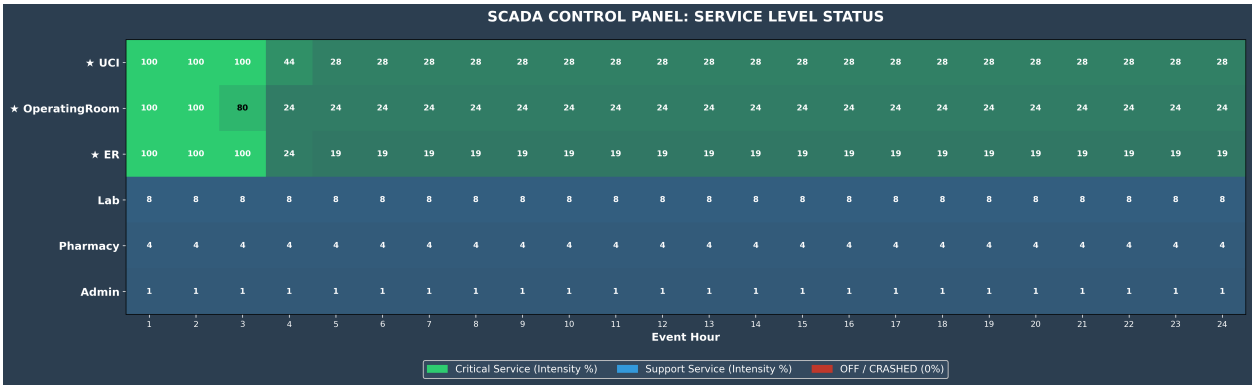


Figure 9: Service Levels - Balanced Strategy (Budget \$20k)

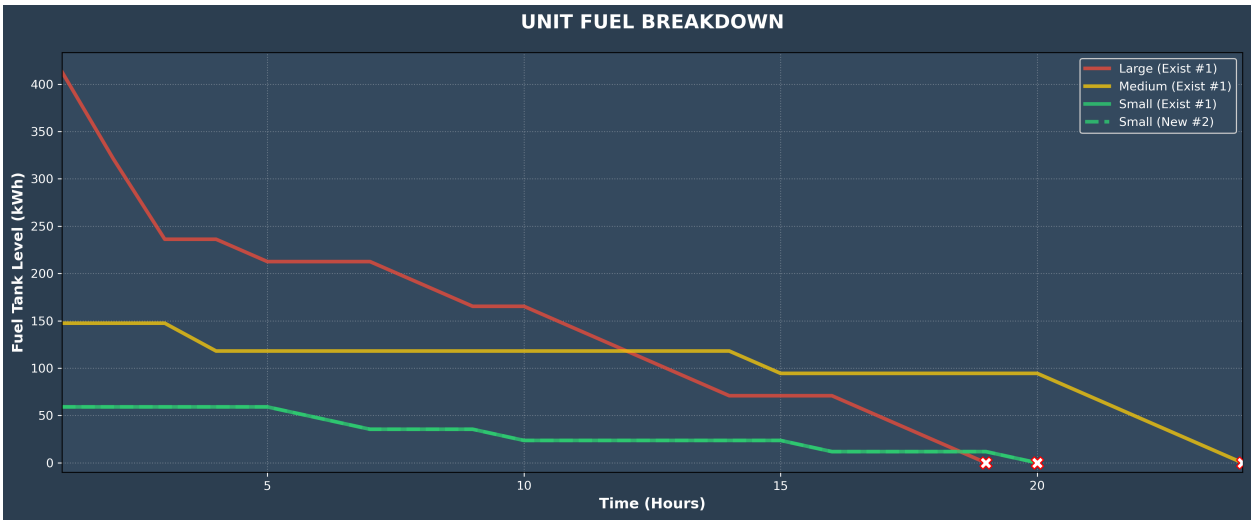


Figure 10: Fuel Depletion - Balanced Strategy (Budget $20k)

6.2 High Budget ($40,000)

Doubling the budget provides the system with the financial capacity to upgrade its investment strategy significantly. Instead of purchasing a Small generator, the model can now afford a 'Medium Generator'.

Observations:

- **Enhanced Power Capacity:** The acquisition of a Medium generator transforms the energy landscape. While the distribution logic (Balanced) remains the same, the *absolute* power available in each hour increases. This allows for higher service levels across the board without sacrificing duration.
- **Improved Efficiency:** As seen in Figure 12, the additional capacity relieves the pressure on the smaller units. The load is distributed more efficiently, and the larger fuel tank of the Medium unit acts as a significant buffer, smoothing out the depletion curve and ensuring a more robust response to the 24-hour demand.

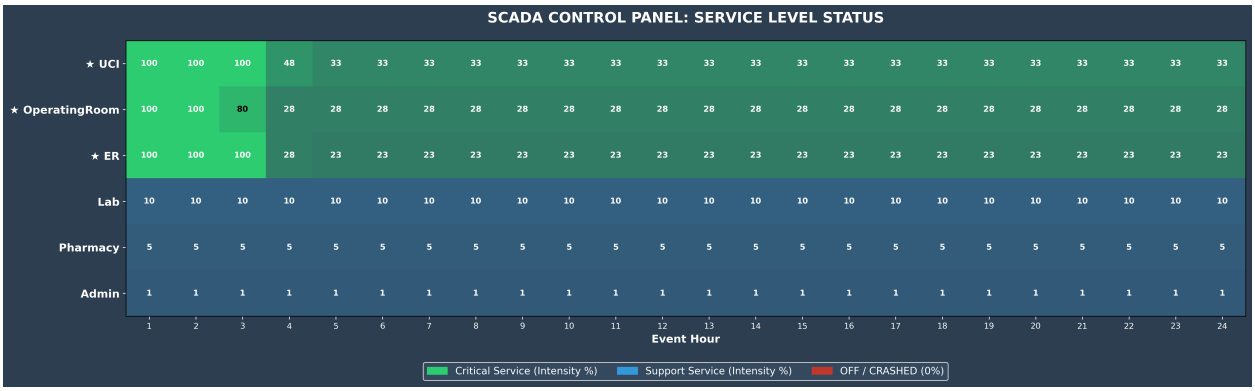


Figure 11: Service Levels - Balanced Strategy (Budget $40k)

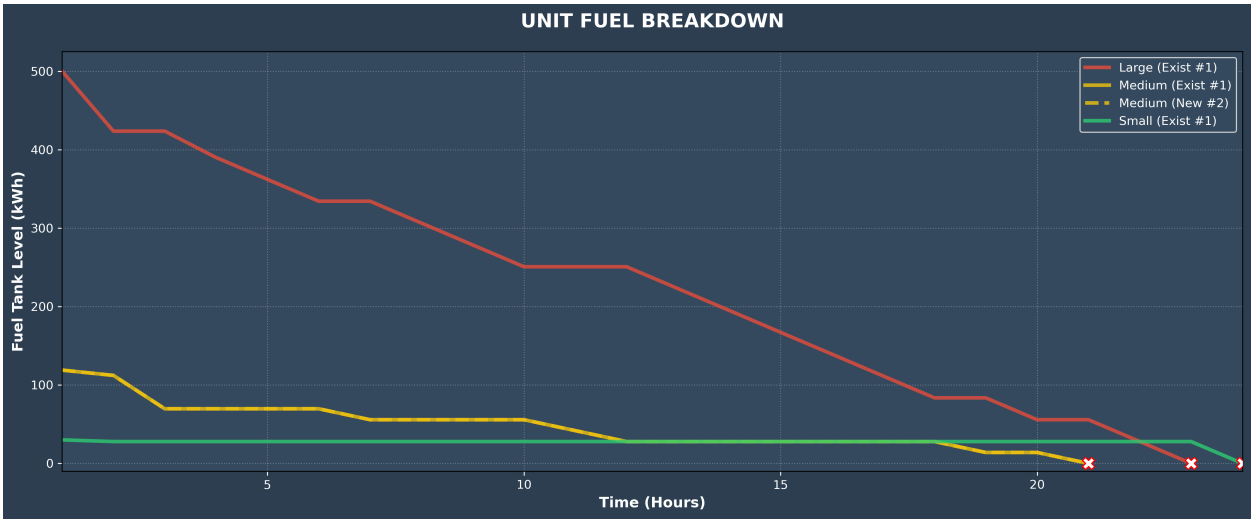


Figure 12: Fuel Depletion - Balanced Strategy (Budget $40k)

7 Long Duration Stress Test (100 Hours)

Finally, we simulate a catastrophic scenario lasting 100 hours (approximately 4 days). For this extreme test, we provide an increased budget of $40,000 but apply a **High Equity** strategy ($\lambda_1 = 0.1, \lambda_2 = 0.9$). The goal is to observe how the system attempts to maintain widespread survivability over such a prolonged period.

7.1 Extreme Resilience ($\lambda_2 = 0.9$) - 100 Hours

In this "survival mode" simulation, the model faces the challenge of stretching finite fuel reserves across a timeline four times longer than the baseline.

Observations:

- **Strategic Sacrifices:** Unlike shorter scenarios where equity kept everyone afloat, the 100-hour horizon forces a hard choice. As seen in Figure 13, the system deliberately sacrifices the **Admin** service (the least critical node) to ensure the survival of the remaining medical departments. This "triage" allows the energy that would have been consumed by Admin to be redistributed, keeping the other lines above zero.
- **Minimal Vital Levels:** To reach the 100th hour, the operational intensity of all active services is significantly reduced. We do not observe high peak values; instead, the system settles into a low-power steady state, providing just enough energy to avoid total collapse.
- **Ultra-Conservative Fuel Usage:** The fuel depletion curve (Figure 14) is extremely gradual. The generators are operated at minimum efficient loads to trickle-feed the hospital grid, ensuring the tanks do not run dry before the final hour.

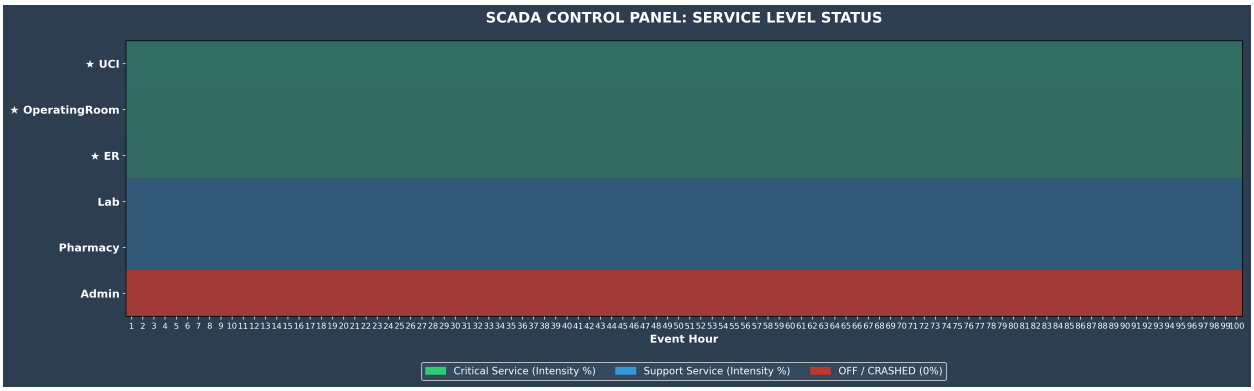


Figure 13: Service Levels (SCADA) - Stress Test (100h, High Equity)

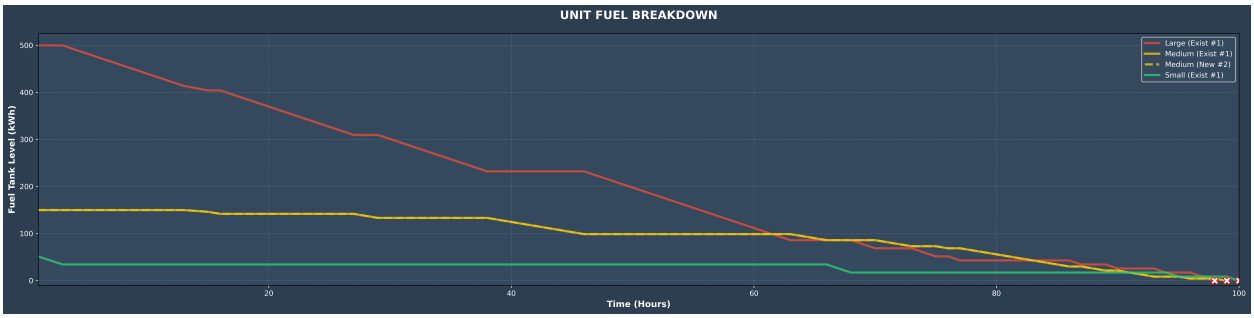


Figure 14: Fuel Depletion - Stress Test (100h, High Equity)

8 Conclusions

This study has developed and analyzed a multi-objective optimization framework for managing hospital energy resilience during blackouts. By simulating a range of scenarios—varying in priority weights (λ), budget constraints, and event duration—we have derived critical insights into the trade-offs between maximizing clinical utility and ensuring equitable service continuity.

8.1 Strategic Trade-offs: Immediacy vs. Persistence

The most significant finding of this research is the fundamental dichotomy between two operational strategies, governed by the objective function weights:

- **The Immediacy Strategy ($\lambda_1 \gg \lambda_2$):** When the model prioritizes Total Utility, it adopts an aggressive "burn-now" approach. Resources are consumed rapidly to power critical services (UCI, Operating Rooms, ER) at maximum capacity in the early hours. While this maximizes the theoretical utility score, it leads to steeper fuel depletion curves and often results in the premature shutdown of support services (Pharmacy, Labs, Admin). This strategy is optimal for short-duration events where survival is not threatened by resource exhaustion.
- **The Persistence Strategy ($\lambda_2 \gg \lambda_1$):** Conversely, prioritizing Equity forces the system into a conservative "survival mode." The model dampens peak performance, maintaining service levels at a lower but uniform standard across all departments. This approach flattens the fuel consumption curve, extending the operational horizon significantly. It proves to be the robust choice for uncertain or long-duration events (48+ hours), where the primary goal shifts from excellence to endurance.

Our 100-hour stress test confirmed that under extreme conditions, a high equity priority is essential to avoid total system collapse, even if it requires strategic sacrifices (such as cutting non-critical administrative loads) to keep the medical core alive.

8.2 The Role of Investment and Capacity

The budget sensitivity analysis revealed that capital investment acts as a direct lever on the system's resilience ceiling:

- **Small vs. Medium Generators:** Under tight budgets ($20,000), the system is forced to rely on existing infrastructure and small, less efficient units. This constrains the "Balanced" strategy ($\lambda_1 = 0.5$), often forcing it to behave more like a scarcity model.
- **Capacity as a Buffer:** Doubling the budget to $40,000 allowed for the acquisition of Medium-sized generators. This did not just add more power; it fundamentally changed the depletion dynamics. The larger fuel tanks and higher output capacity provided a buffer that allowed the system to satisfy both utility and equity objectives simultaneously for longer periods, smoothing out the trade-offs that are starkly visible in lower-budget scenarios.

8.3 Final Recommendations

Based on the quantitative results, we propose the following guidelines for hospital emergency planning:

1. **Dynamic Protocols:** Operational protocols should not be static. Hospitals should adopt a dynamic response plan that shifts priority from Utility (Immediacy) to Equity (Persistence) as the estimated duration of an outage increases.
2. **Investment Efficiency:** For facility upgrades, investing in medium-tier generation capacity offers a sweet spot between cost and resilience, providing the necessary fuel autonomy to support a balanced strategy during 24-hour events.
3. **Load Shedding Hierarchies:** The model demonstrates the value of pre-planned load shedding. Identifying non-critical nodes (like Administration) that can be sacrificed early allows for the redistribution of energy to life-saving units, a practice that is mathematically validated by our 100-hour simulation.

In conclusion, while no single strategy is optimal for every possible disaster, this optimization framework provides decision-makers with the quantitative evidence needed to navigate the complex ethical and logistical choices inherent in healthcare crisis management.